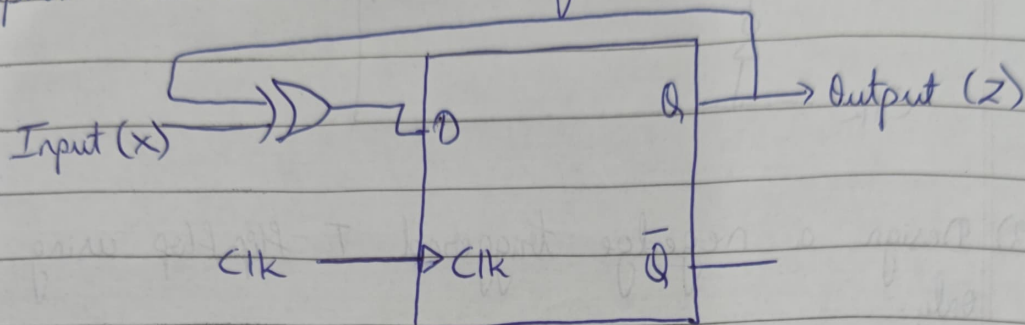


Day 204 Sequential Assignment

2) Explain the behaviour of the given circuit



Solⁿ From the above $D = X \oplus Q$ and $Q_{next} = D$

Q	X	$D = X \oplus Q$	Q_{next}
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

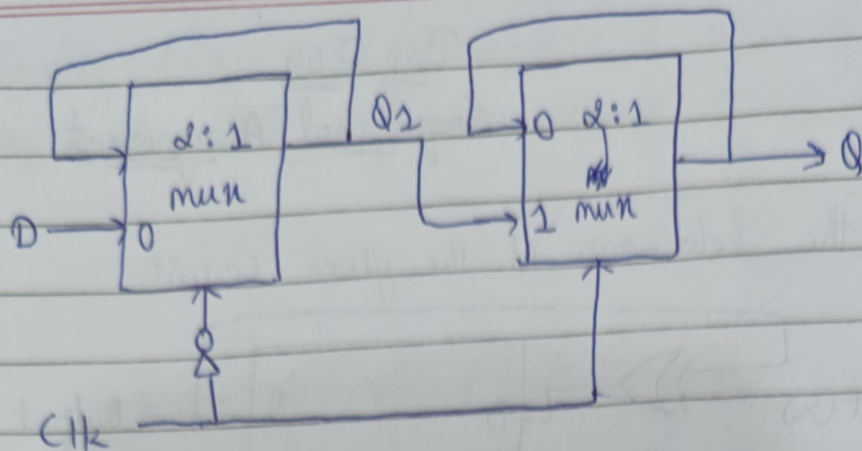
Comparing input X and the next state Q_{next} ;

when $X = 0$ $Q_{next} = Q$ and

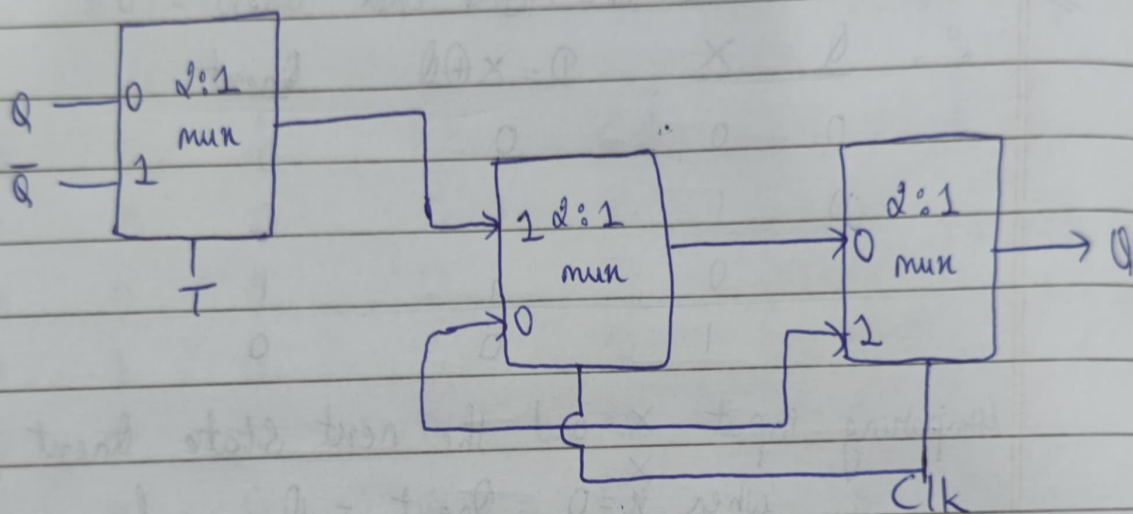
when $X = 1$ $Q_{next} = \bar{Q}$

Hence, the behaviour of the given circuit is a T-Flip-Flop

2) Design a posedge triggered D Flip Flop using 2:1 Muxes only

Soln

3) Design a negedge triggered T flip flop using 2:1 Muxes only.

Soln

4) Convert by adding external gates:

- A D flip flop to a J-K flip flop
- A T flip flop to a D flip flop

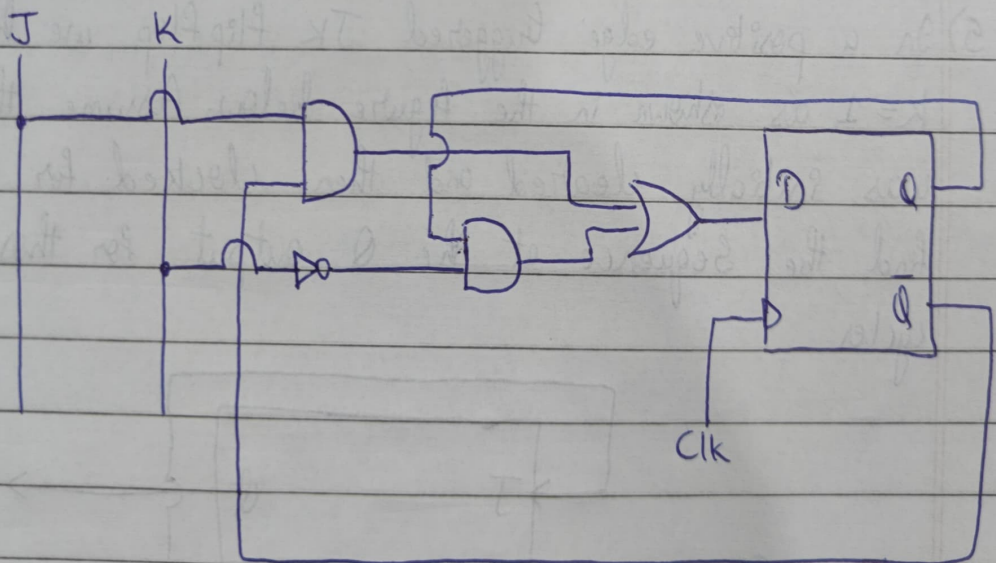
Soln

a) A D Flip Flop to a J-K Flip Flop

<u>J</u>	<u>K</u>	<u>Q (Present State)</u>	<u>Q_{next}</u>
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

J \ KQ	00	01	11	10
0	0	1	0	0
1	1	1	0	1

$$D = Q_{next} = J\bar{Q} + \bar{K}Q$$

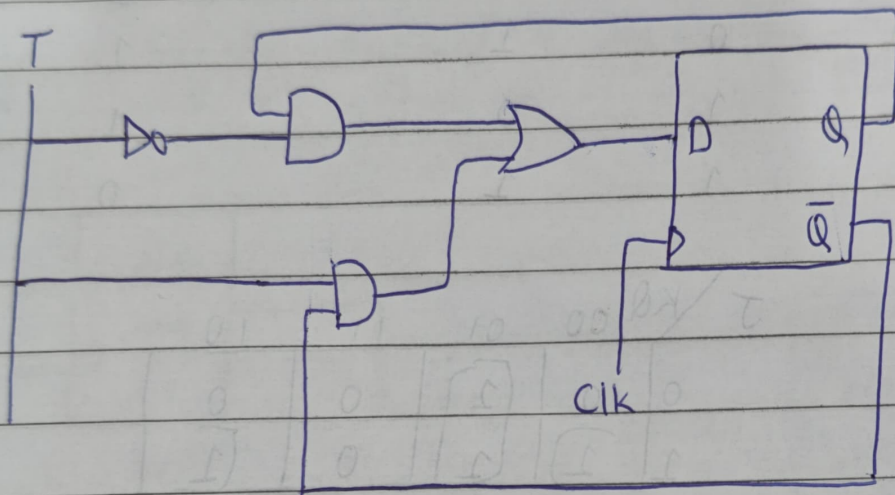


b) A T flip flop to a D flip flop

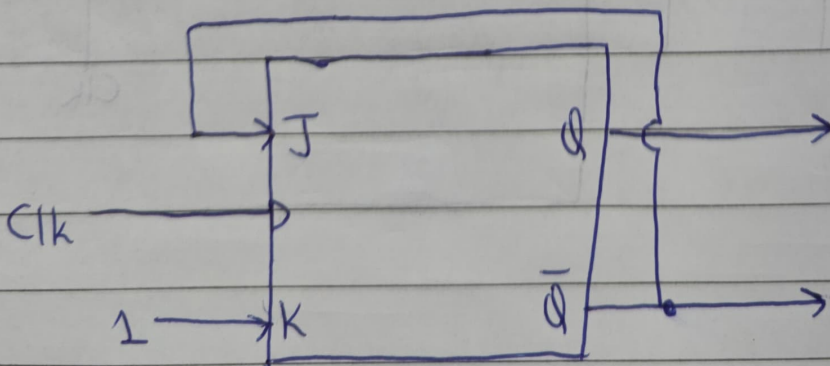
<u>T</u>	<u>Q(Present)</u>	<u>Q_{next}</u>
0	0	0
0	1	1
1	0	1
1	1	0

T \ Q	0	1
0	0	1
1	1	0

$$Q_{next} = \bar{T}Q + T\bar{Q}$$



5) In a positive edge triggered JK flip flop, we have $J = Q'$ and $K = 1$ as shown in the figure below. Assume the flip flop was initially cleared and then clocked for 6 clock pulses. Find the sequence at the Q output for those 6 clock cycles.



Sam Positive edge

	J	K	Q	$\bar{Q}$
1st	1	1	1	0
2nd	0	1	0	1
3rd	1	1	1	0
4th	0	1	0	1
5th	1	1	1	0
6th	0	1	0	1

∴ The Sequence Q = 1, 0, 1, 0, 1, 0

0	0	1	0	0
X	X	0	1	1

1	1	0	1	0
X	X	1	0	1